Introduction to Photosynthesis

How plants make food from sunlight

- Photosynthesis converts light energy into chemical energy stored in glucose
- It occurs in the chloroplasts of plant cells, which contain chlorophyll
- Photosynthesis is essential for almost all life on Earth

The Raw Materials

What does a plant need?

- Carbon dioxide (CO_2) -- absorbed through tiny pores called stomata
- Water (H_2O) -- drawn up from the roots through the stem
- Light energy -- captured by the green pigment chlorophyll

The Two Stages

Light reactions and the Calvin cycle

- Light reactions: in the thylakoid membranes -- split water, produce ATP
- Calvin cycle: in the stroma -- uses ATP to fix CO_2 into sugars
- Together they convert light energy into chemical energy as glucose

The Overall Equation

$6CO_2 + 6H_2O$ + light --> $C_6H_{12}O_6 + 6O_2$

- Six molecules of CO_2 plus six of water plus light yields one glucose molecule
- Oxygen is released as a by-product -- the source of the air we breathe
- One glucose molecule stores about 686 kcal of chemical energy

Why It Matters

Photosynthesis underpins all food chains

- Plants are the primary producers at the base of every food web
- Photosynthesis removes CO_2 from the atmosphere, regulating climate
- Understanding it is key to improving crop yields and renewable energy